

Yu Li

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Education

George Washington University	Washington, D.C.
<i>Ph.D. Candidate in Electrical and Computer Engineering, GPA: 4.0/4.0</i>	<i>2025.08 – 2029.05</i>
Advisor: Prof. Tian Lan 📧; Work with Prof. Zhengling Qi 📧	<i>(expected)</i>
Wuhan University, Hongyi Honor College	Wuhan, China
<i>B.Eng. in Microelectronics Science and Engineering, GPA: 3.87/4.0</i>	<i>2021.09 – 2025.06</i>

Research Highlights

- **LLM Reasoning:** LLMs alignment and policy optimization. [\[P1\]](#)[\[P3\]](#)[\[P4\]](#)[\[P5\]](#)
- **Agentic RL:** RL for agent tasks and multi-agent systems. [\[P2\]](#)[\[C2\]](#)[\[C5\]](#)
- **Efficient AI:** LLMs efficient training and inference. [\[C1\]](#)[\[C6\]](#)[\[C7\]](#)
- **Generative Models:** Diffusion-based generation. [\[C3\]](#)[\[J2\]](#)

Industry Experience

ByteDance Inc.	San Jose, CA
<i>Machine Learning Scientist Intern, Tiktok Recommendation Core</i>	<i>2026.05 – 2026.08</i>
◦ Worked on production-scale video data for TikTok’s recommendation system. ◦ Focused on the ranking stage, developing LLM-based ranking models for candidate scoring. ◦ Integrated content understanding signals into ranking features within a cross-functional team.	

Preprints & Under Review

P=Preprint

- [P1] **OPPO: Bayesian Value Recursion for Token-Level Credit Assignment in LLM Reasoning** 📧
Yu Li, Rui Miao, Tian Lan, Zhengling Qi
- [P2] **ARISE: Agent Reasoning with Intrinsic Skill Evolution in Hierarchical Reinforcement Learning** 📧
Yu Li, Rui Miao, Zhengling Qi, Tian Lan
- [P3] **Right Meets Wrong: Bilateral Context Conditioning with Reward-Confidence Correction for GRPO** 📧
Yu Li, Tian Lan, Zhengling Qi
- [P4] **INSPO: Unlocking Intrinsic Self-Reflection for LLM Preference Optimization** 📧
Yu Li, Tian Lan, Zhengling Qi
AI with Recursive Self-Improvement Workshop at ICLR, 2026
- [P5] **Hindsight Self-Distillation: Demonstrating to the Teacher for On-Policy LLM Reasoning** 📧
Yu Li, Shu Hong, Tian Lan
Under review at EMNLP 2026, OA: 3.33/5

Publications

C=Conference, J=Journal, †=Equal Contribution

- [C1] **MomentKV: Closing the Directional Gap in KV Cache Eviction for Long-Context Inference**
Yu Li, Binxu Li, Tian Lan
Conference on Language Modeling (COLM), 2026
- [C2] **Reason in Chains, Learn in Trees: Self-Rectification and Grafting for Multi-turn Agent Policy Optimization**
Yu Li, Sizhe Tang, Tian Lan
Findings of the Association for Computational Linguistics (ACL), 2026
- [C3] **CRAFT-LoRA: Content-Style Personalization via Rank-Constrained Adaptation and Training-Free Fusion**
Yu Li, Yujun Cai, Chi Zhang
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026
- [C4] **KG-SAM: Encoding Structural Constraints into Segment Anything Models via Probabilistic Graphical Models**
Yu Li, Da Chang, Xi Xiao
IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) [Oral], 2026
- [C5] **ACDZero: Graph-Embedding-Based Tree Search for Mastering Automated Cyber Defense**
Yu Li†, Sizhe Tang†, Rongqian Chen, Fei Xu Yu, Guangyu Jiang, Mahdi Imani, Nathaniel D. Bastian, Tian Lan
Intelligent Cloud Computing and Networking Workshop at INFOCOM, 2026
- [C6] **Calibrating and Rotating: A Unified Framework for Weight Conditioning in PEFT**
Da Chang, Peng Xue, Yu Li, Yongxiang Liu, Pengxiang Xu, Shixun Zhang
The 40th Annual AAAI Conference on Artificial Intelligence (AAAI), 2026
- [C7] **Mixed Text Recognition with Efficient Parameter Fine-Tuning and Transformer**
Yu Li†, Da Chang†
International Conference on Neural Information Processing (ICONIP), 2024
- [J1] **Dual branch SAM-Transformer Fusion Network for Accurate Breast Ultrasound Image Segmentation**
Yu Li, Jin Huang, Du Wang, Liye Mei, Cheng Lei *et al.*
Medical Physics, 2025
- [J2] **SfMDiffusion: Self-supervised Monocular Depth Estimation in Endoscopy Based on Diffusion Models**
Yu Li, Da Chang, Jin Huang, Du Wang, Liye Mei, Cheng Lei *et al.*
International Journal of Computer Assisted Radiology and Surgery, 2025

Selected Awards & Honors

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- **ICML 2026 Gold Reviewer**, top reviewers recognition. 2026
 - **Innova International Exchange Scholarship**, 6 recipients university-wide, ¥70,000. 2024
 - **Innova Excellence Scholarship**, Top 3%, ¥10,000, twice. 2023, 2024
 - **First-Class Scholarship**, Top 5%, ¥3,000, three times. 2022, 2023, 2024

Academic Services

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- **Conference Reviewer:** {NeurIPS, COLM, MM, ICML, ICLR, AAAI, ICASSP}'2026
 - **Journal Reviewer:** TMLR, TPAMI, ToN, Neurocomputing